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1.0 Introduction

1.1 Project Overview

The goal of this project is to investigate and use data mining methods, particularly classification algorithms, to extract useful information from the chosen dataset. The dataset includes structured data related to a certain business domain that was previously used in an earlier assignment. For each of the three unique business insights, the project's main goal is to create, train and assess a unique categorization algorithm. Data preparation, model training, assessment and interpretation are all steps in the process. The project shows how several machine learning models may be used to address practical issues by matching each algorithm with a specific business topic. A comparison evaluation of the model's performances and how effectively it aids in corporate decision-making are included in the final product.

1.2 Objectives

- To apply one classification algorithm for each business insight, tailored to the nature of the problem
- To perform necessary data preprocessing, including cleaning, transformation and sampling to ensure data quality
- To develop, train and evaluate three classification models using suitable performance metrics
- To interpret the outcomes of each model and explain how the results provide value or recommendations for business decision-making
- To document the full process in a clear, technical report supported with visualizations and analysis

1.3 Business Questions/Insights Targeted

1.3.1 Prevalence of Diabetes in the Population

Analysis of the BRFSS 2015 dataset shows that around 29.7% of respondents are diagnosed with diabetes or pre-diabetes. This highlights a significant public health concern. Early detection and lifestyle-based prevention strategies are essential to reduce future cases and healthcare costs. This insight supports the development of targeted health interventions and resource allocation for high-risk groups.

1.3.2 Impact of Lifestyle Factors on Health Outcomes

Lifestyle behaviors such as smoking, physical activity, and diet significantly influence health outcomes. Analysis shows that 44.3% of respondents smoke, while 75.6% engage in regular physical activity and 63.4% consume fruits daily—highlighting both healthy habits and areas of concern. These insights underscore the need for targeted public health interventions, including smoking cessation, fitness programs, and nutrition education, to reduce chronic disease risks and improve overall population health.

1.3.3 Predictive Modeling for At-Risk Individuals

Analysis shows that key health features such as high blood pressure, cholesterol levels, BMI, smoking status, and exercise frequency can effectively identify individuals at high risk of diabetes. Although the overall accuracy is slightly lower compared to other methods, this approach demonstrates better sensitivity in detecting potential high-risk cases, making it valuable for early identification. Applying this predictive model in healthcare or corporate wellness programs allows for immediate risk assessment and personalized health recommendations after users complete health questionnaires, thereby improving screening efficiency and optimizing preventive care.

2.0 Dataset Description

2.1 Source of Dataset

The Behavioral Risk Factor Surveillance System (BRFSS) 2015, a nationwide health survey carried out by the Centers for Disease Control and Prevention (CDC) in the United States, served as the dataset for this study. Through telephone interviews, it gathers data on chronic health issues, health related risk behaviors and utilization of preventative services.

2.2 Description of Features

The original dataset, `diabetes_012_health_indicators_BRFSS2015.csv`, includes over 400,000 records and the 22 features in these following 6 categories: health status, lifestyle, healthcare access, medical history, demographics and target.

2.3 Target Variable

There is 2-class categorical variable:

0 – No diabetes

1 – Diabetes

2.4 Summary Statistics

The cleaned dataset (`dmafterprocess.csv`) retains all original features but has undergone the following cleaning steps which is removal of missing or inconsistent data, standardization or transformation of certain variables and class balancing to improve model performance.

3.0 Data Preprocessing

Our group used KNIME Analytics Platform to perform data preprocessing

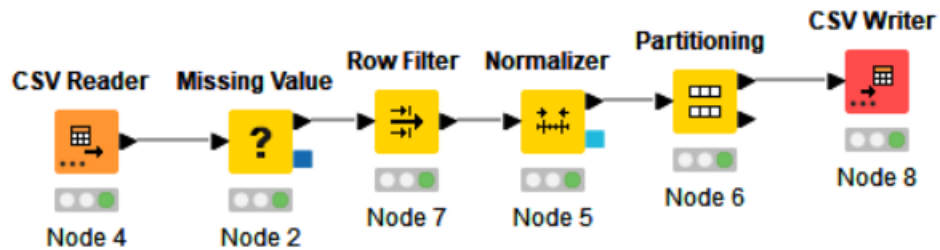


Figure 1: Data Preprocessing using KNIME Analytics Platform

3.1 Missing Value Handling

No missing values were present in the original dataset (diabetes_012_health_indicators_BRFSS2015.csv), so no imputation or removal was necessary. This was confirmed during the initial data inspection phase.

3.2 Row Filtering

Some entries had illogical or extreme values, particularly in the BMI column, with values reaching up to 98, which are unrealistic. To maintain data quality, all rows with BMI > 60 were removed. These entries appeared to be erroneous or random, so filtering helped avoid model distortion.

Filter

Criterion 1

Filter column: BMI

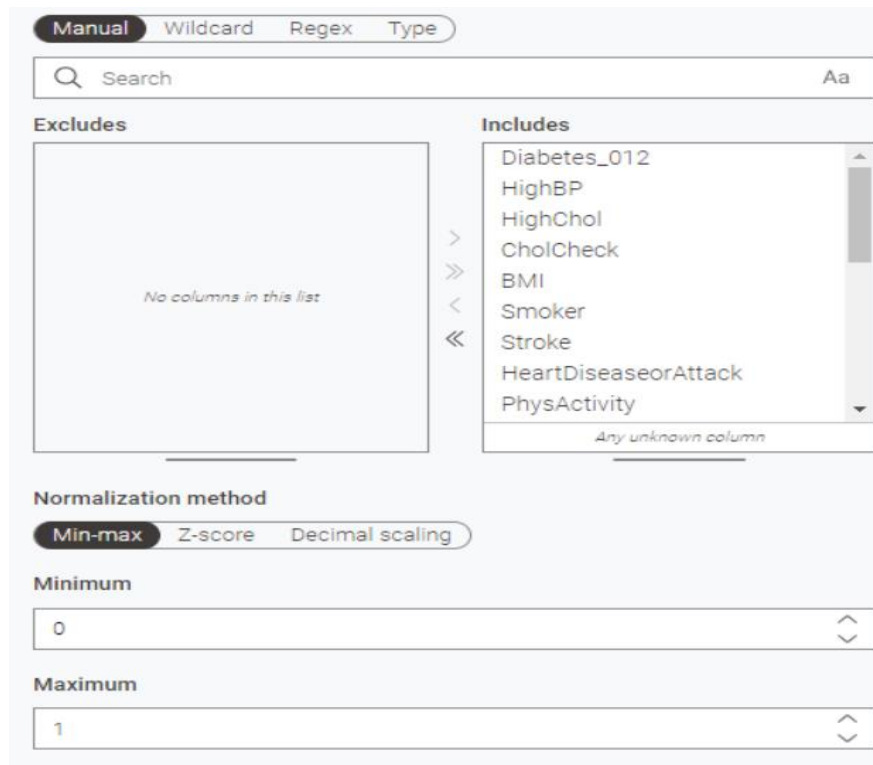
Operator: Less than

Value: 60

Figure 2: Row filtering in KNIME Analytics Platform

3.3 Normalization

All numerical features (such as BMI, Age, Income, GenHlth, etc) were normalized to a [0, 1] range using Min-Max scaling. This transformation ensures that all values are on a consistent scale, which improves the performance of models like KNN and neural networks.



The screenshot shows the configuration window for the 'Normalization' widget in KNIME. At the top, there are four tabs: 'Manual' (selected), 'Wildcard', 'Regex', and 'Type'. Below the tabs is a search bar with a magnifying glass icon and the text 'Search', followed by a text input field containing 'Aa'. The main area is divided into two columns: 'Excludes' and 'Includes'. The 'Excludes' column is empty and contains the text 'No columns in this list'. The 'Includes' column contains a list of features: 'Diabetes_012', 'HighBP', 'HighChol', 'CholCheck', 'BMI', 'Smoker', 'Stroke', 'HeartDiseaseorAttack', and 'PhysActivity'. To the left of this list are four arrow buttons: '>', '>>', '<', and '<<'. At the bottom of the 'Includes' list is the text 'Any unknown column'. Below the columns, there is a section for 'Normalization method' with three buttons: 'Min-max' (selected), 'Z-score', and 'Decimal scaling'. At the bottom, there are two input fields: 'Minimum' with the value '0' and 'Maximum' with the value '1'. Both input fields have up and down arrow buttons on the right side.

Figure 3: Normalization in KNIME Analytics Platform

3.4 Data Partitioning

The preprocessed dataset represents an 80% training subset of the full dataset. A random 80/20 split was applied in which 80% of data for training and modeling and 20% reserved for testing. This split was stratified to ensure proportional representation of all classes in the target variable Diabetes_012.

The screenshot shows the 'First partition' tab in the KNIME Analytics Platform. The title bar includes 'First partition', 'Flow Variables', 'Job Manager Selection', and 'Memory Policy'. The main area is titled 'Choose size of first partition'. It contains several radio buttons: 'Absolute' (unselected), 'Relative[%]' (selected), 'Take from top' (unselected), 'Linear sampling' (unselected), 'Draw randomly' (selected), 'Stratified sampling' (unselected), and 'Use random seed' (unchecked). To the right of 'Absolute' is a text box with '100'. To the right of 'Relative[%]' is a text box with '80'. To the right of 'Stratified sampling' is a dropdown arrow. To the right of 'Use random seed' is a text box with '1,749,434,588,8'.

Figure 4: Data Partitioning in KNIME Analytics Platform

3.5 Data Discretization

Although data discretization is typically used to convert continuous features into categorical bins, it was not applied in this project because the dataset primarily consists of binary or already scaled numerical features. Also, all continuous features fall within $[0, 1]$ after normalization.

4.0 Model Development

4.1 Overview of Selected Classification Algorithms

Algorithm 1: XGBoost

XGBoost is an optimized distributed gradient boosting library. It is also a type of ensemble learning method that provides a parallel tree boosting (also known as GBDT) that uses decision trees as its base learners combining them sequentially to improve the model's performance. It is known for its speed, efficiency and ability to scale well with large datasets.

Algorithm 2: Random Forest

To provide predictions, Random Forest, an ensemble classification technique, aggregates the majority vote from several decision trees that are constructed during training. By adding randomization to both the feature selection and each node split and the data sampling procedure (by bootstrapping), it improves generalization and lessens overfitting. The resilience of the model is increased because of the inherent randomization, which guarantees that every tree is marginally unique. Random Forest is a strong and interpretable algorithm for real-world applications because it works well on high-dimensional datasets, is noise-resistant and offers insightful information about feature relevance.

Algorithm 3: K-Nearest Neighbours (KNN)

K-Nearest Neighbour (KNN) is a non-parametric, instance-based learning algorithm used for classification tasks. It classifies new data points by calculating the distance (usually Euclidean) to all training data points and selecting the most common class among the k-nearest neighbour.

4.2 Implementation Tools and Libraries Used

Implementation Tools

- KNIME: Data preprocessing
- Python: Core programming language used for scripting and data analysis

Libraries

- Pandas: For data loading, filtering, transformation and summary statistics
- Sklearn: The main library used for data preprocessing, classification models and evaluation metrics
- Matplotlib & Seaborn: For data visualization and plotting graphs like feature distributions, correlation heatmaps and evaluation curves

- Google Colab: As the development environment for code writing, execution and visualization

4.3 Model Training & Tuning

Each of the three classifiers (XGBoost, Random Forest and KNN) was trained using the 80% training subset of the cleaned dataset. The models were tuned using basic hyperparameter settings and validated using the remaining 20% test data. The training process for all models began with normalization of the dataset to ensure consistent input scales, which is particularly important for distance-based algorithms like KNN. The data was then split into training (80%) and test (20%) sets using stratified sampling to preserve the original class distribution. Each classification model was trained using the specific set of features relevant to its corresponding business insight. Model performance was subsequently evaluated using key metrics such as accuracy, precision, recall, and F1-score to assess both overall effectiveness and the ability to correctly identify cases of interest.

Hyperparameter Tuning Summary

The table below summarizes the key hyperparameters tuned for each classification model. Tuning was performed using manual testing and cross-validation to find the best trade-off between accuracy and generalization.

Model	Key Parameters Tuned	Tuning Approach
XGBoost	n_estimators = 100, max_depth = 6, learning_rate = 0.1, scale_pos_weight	Empirical tuning using validation accuracy and class imbalance handling
Random Forest	n_estimators = 100, max_depth = None, class_weight = 'balanced'	Manual tuning based on performance on validation set
KNN	n_neighbors = 5, weights = 'uniform', metric = 'euclidean'	Tested multiple k values; chose best based on cross-validation accuracy

5.0 Model Evaluation & Comparison

5.1 Performance Metrics (Accuracy, Precision, Recall, F1-score)

Model	Class	Precision	Recall	F1-score	Accuracy
XGBoost	0	0.88	0.98	0.93	0.86
	1	0.53	0.17	0.25	
Random Forest	0	0.86	0.96	0.91	0.84
	1	0.48	0.19	0.27	
KNN	0	0.86	0.95	0.90	0.83
	1	0.39	0.19	0.25	

5.2 Confusion Matrix & Evaluation Charts

XGBoost

Confusion Matrix

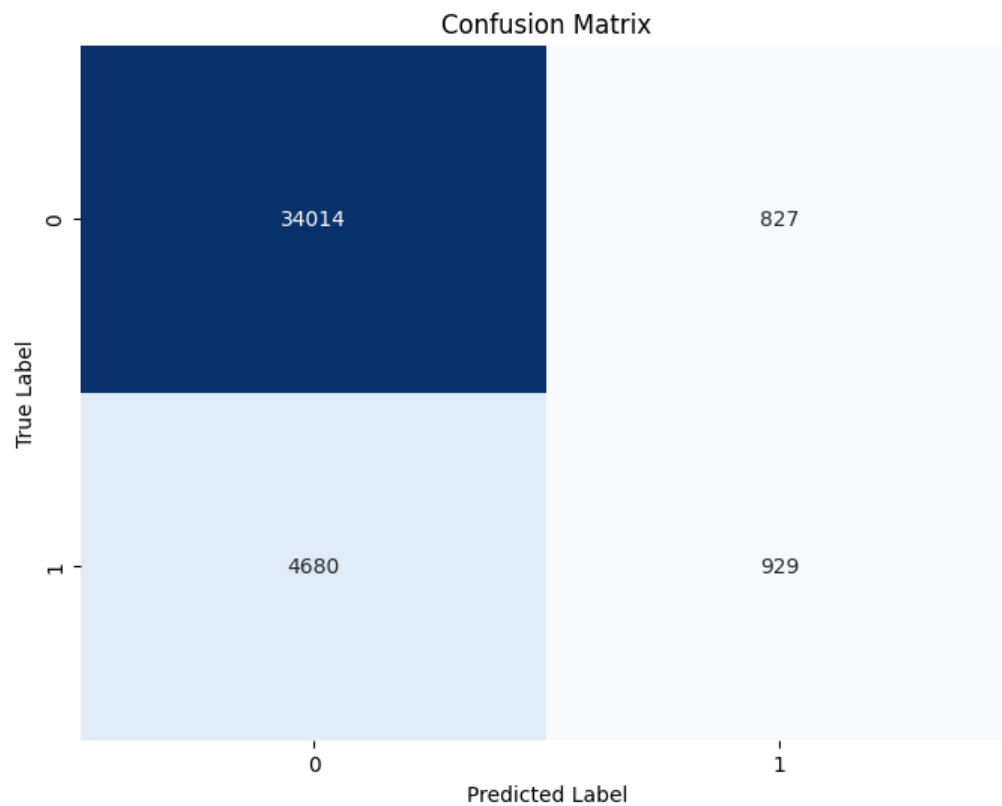


Figure 5: Confusion matrix using XGBoost

Evaluation Charts:

1. ROC Curve

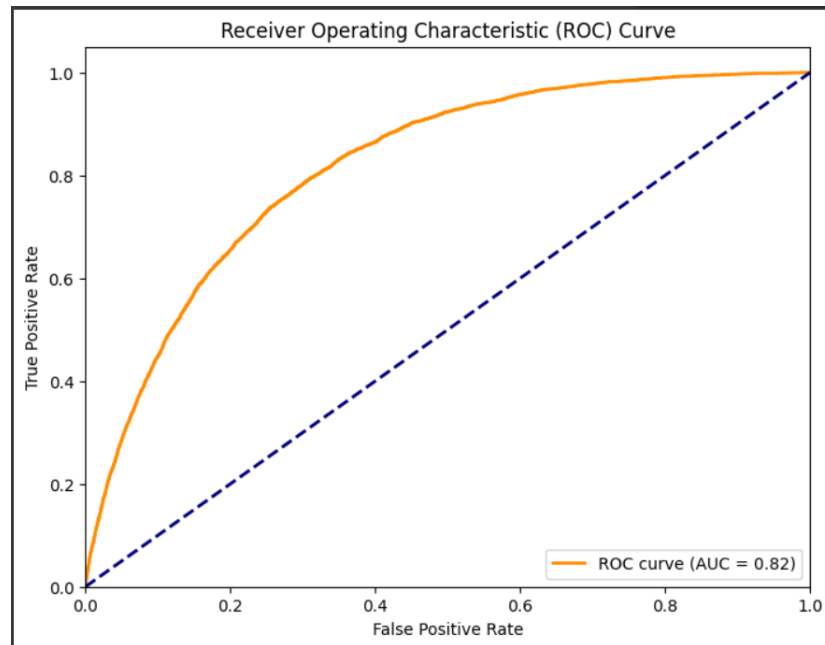


Figure 6: ROC Curve for XGBoost

2. Precision-Recall Curve

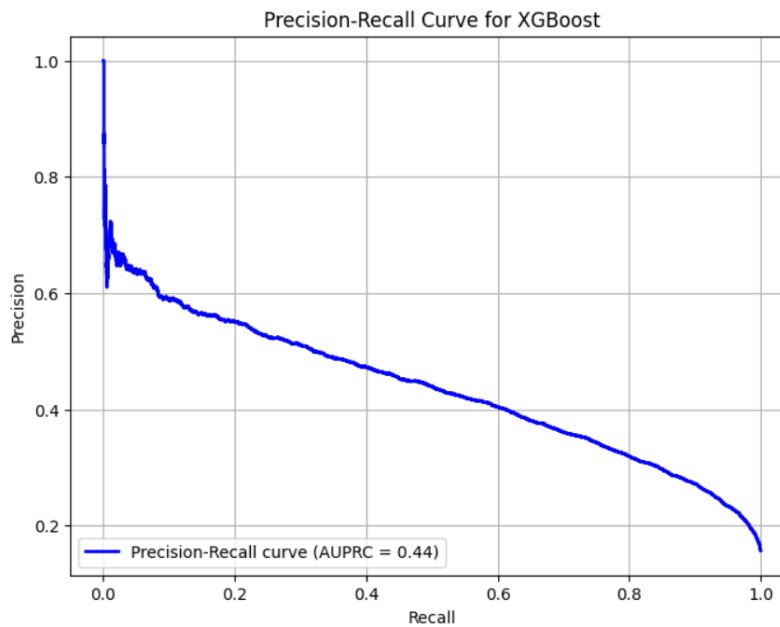


Figure 7: Precision-Recall Curve for XGBoost

3. Classification Reports using Bar Chart

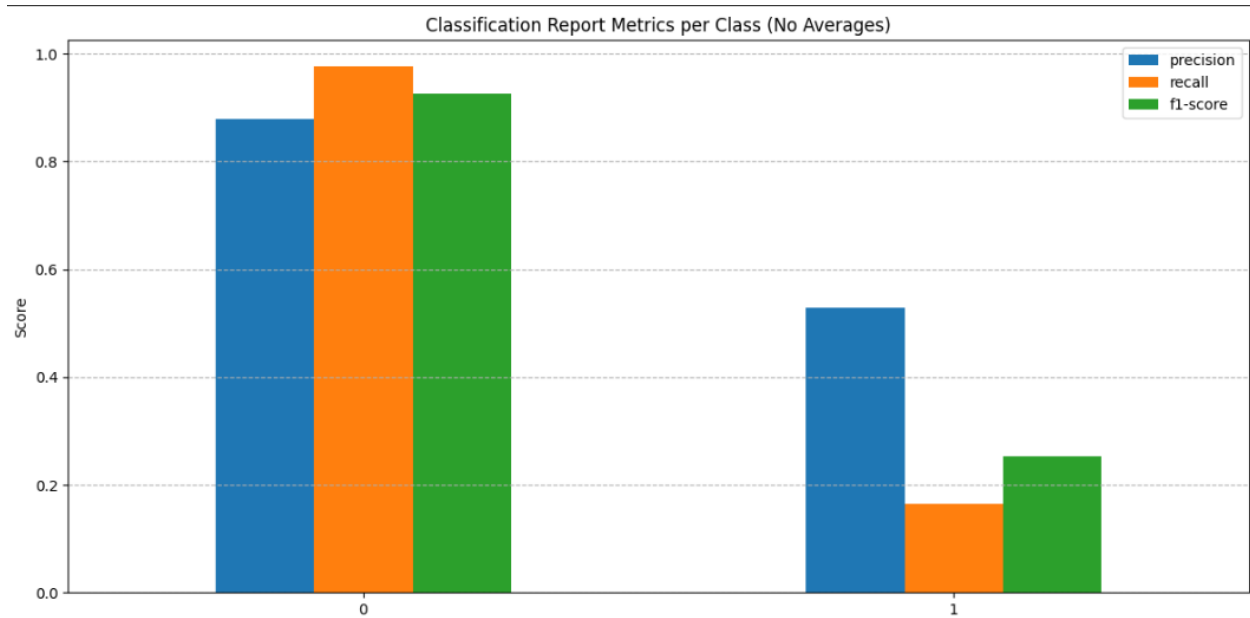


Figure 8 : Classification Report for XGBoost

This bar chart visualizes the precision, recall, and f1-score for each of the classes (0 for no diabetes, 1 for diabetes), as well as the macro and weighted averages.

Class 0 (No Diabetes): We can see that the model has high precision, recall, and f1-score for this class. This means that when the model predicts someone does not have diabetes, it's usually correct (high precision), and it's able to identify most of the people who actually don't have diabetes (high recall).

Class 1 (Diabetes): The metrics for this class are significantly lower, particularly recall and f1-score. This indicates that the model is not as good at identifying individuals who actually have diabetes. The lower precision means that when the model predicts someone has diabetes, it's wronged a significant portion of the time. The low recall means it misses many individuals who actually have diabetes.

Random Forest

Confusion Matrix

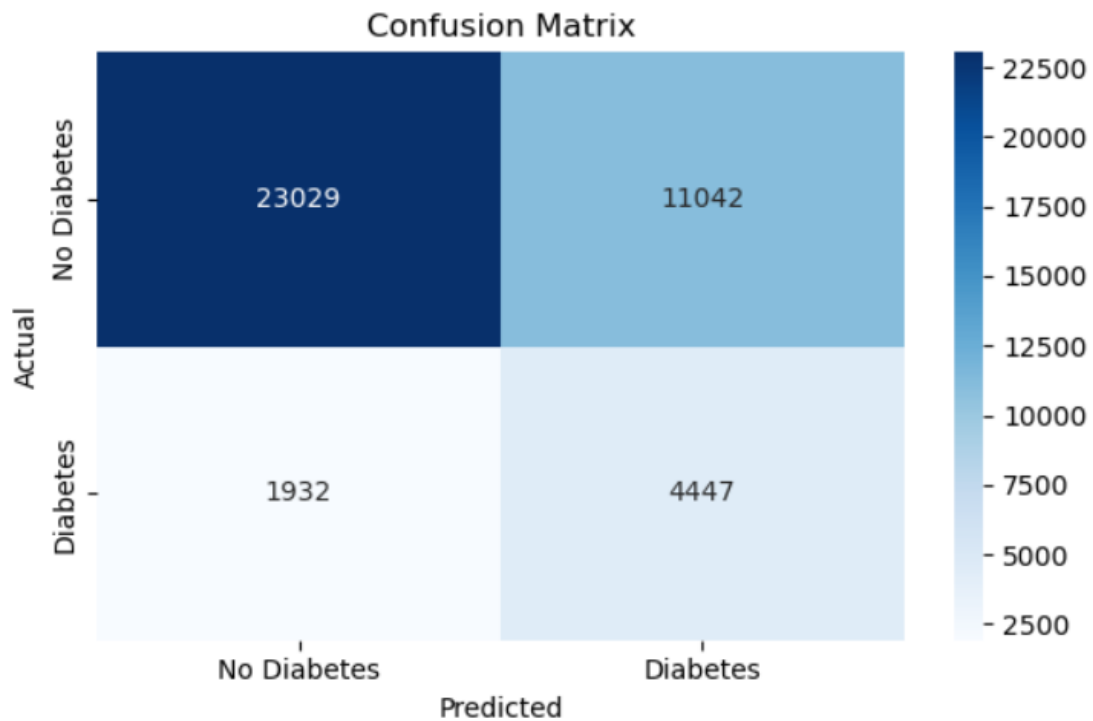


Figure 9 : Confusion Matrix using Random Forest

Evaluation charts:

1. ROC Curve

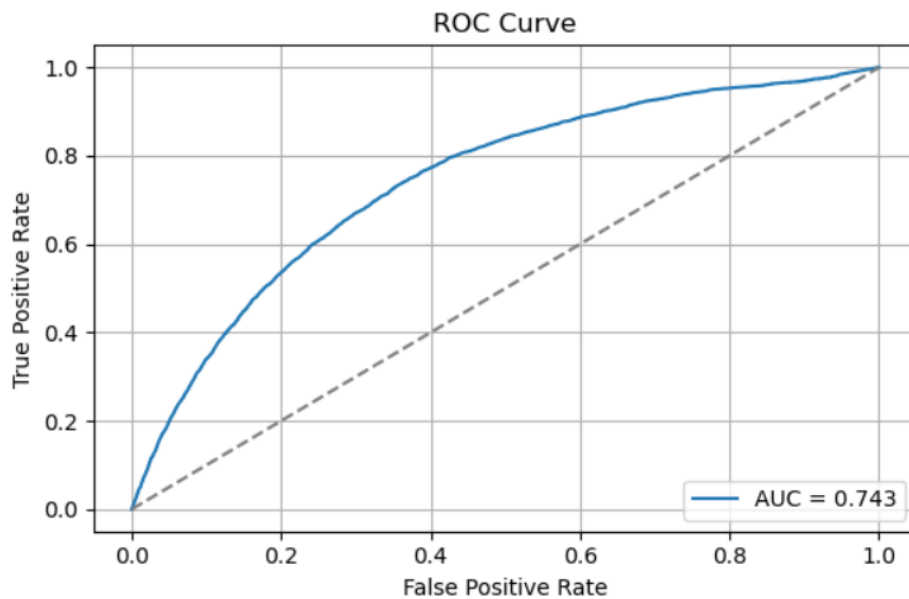


Figure 10: ROC Curve using Random Forest

2. Precision-Recall Curve

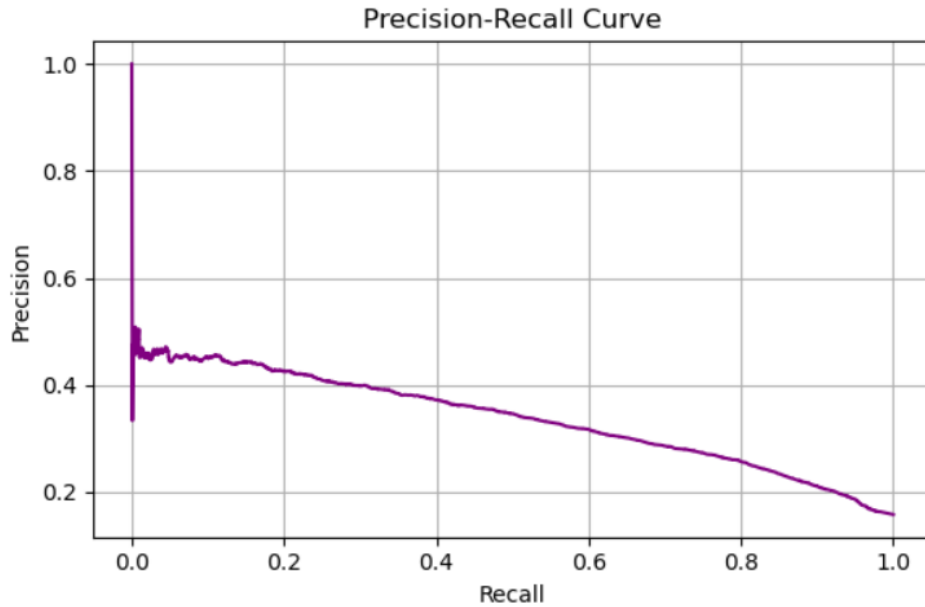


Figure 11: Precision-Recall Curve using Random Forest

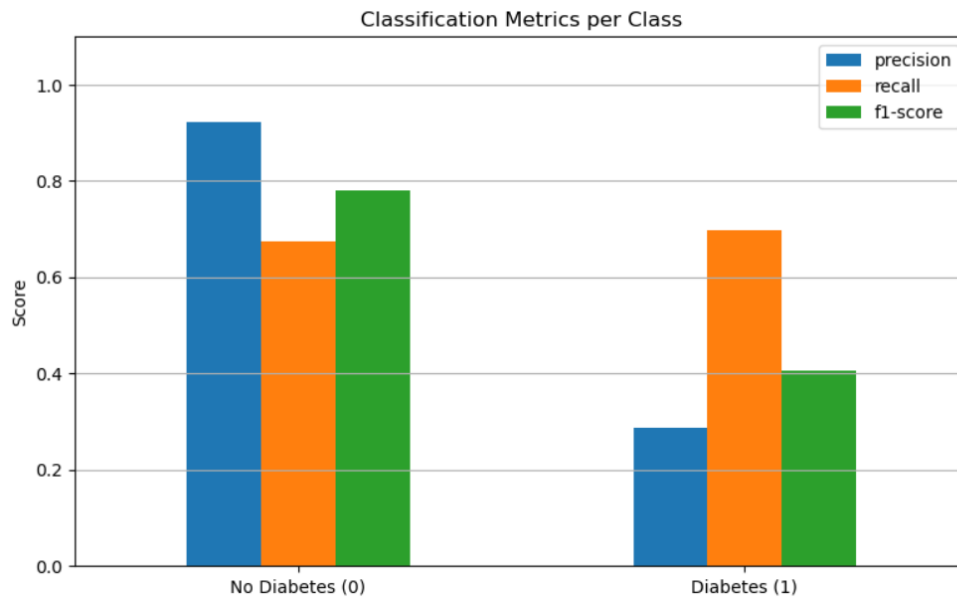


Figure 12: Bar Chart - Classification Metrics per Class using Random Forest

The bar chart provides a clear comparison of the model's performance across two classes: Class 0 (No Diabetes) and Class 1 (Diabetes), based on three key metrics which is precision, recall and F1-score. These metrics help evaluate how well the Random Forest classifier, when trained exclusively on lifestyle-related features such as smoking, physical activity and diet, can distinguish between diabetic and non-diabetic individuals.

For Class 0 (No Diabetes), the model shows consistently high performance. The high precision means that the model is typically right when it predicts that a person does not have diabetes. The

model effectively detects the great majority of people who are not diabetic, as evidenced by its high recall. This class's high F1-score indicates that the model is very successful at identifying non-diabetic cases because it strikes a solid balance between precision and recall. Given that Class 0 comprises most of the sample, this impressive performance is to be expected.

In contrast, the performance for Class 1 (Diabetes) is slightly lower. The recall is noticeably reduced, but the precision is still rather high, indicating that any of the people who were predicted to have diabetes are in fact affected. This means that the model misses several diabetic cases, likely due to the limited scope of using lifestyle features alone, which may not fully capture all risk factors associated with diabetes. As a result, the F1-score for this class reflects a moderate performance, suggesting room for improvement in detecting diabetic individuals based on behavior-related inputs.

Overall, the bar chart shows that the model performs mediocly well in recognizing cases of diabetes and is quite dependable in identifying people who do not have the disease. These variations, which are graphically depicted by the bar heights, offer important information of integrating a broader set of features in future models to improve detection of at-risk individuals.

KNN

Confusion Matrix

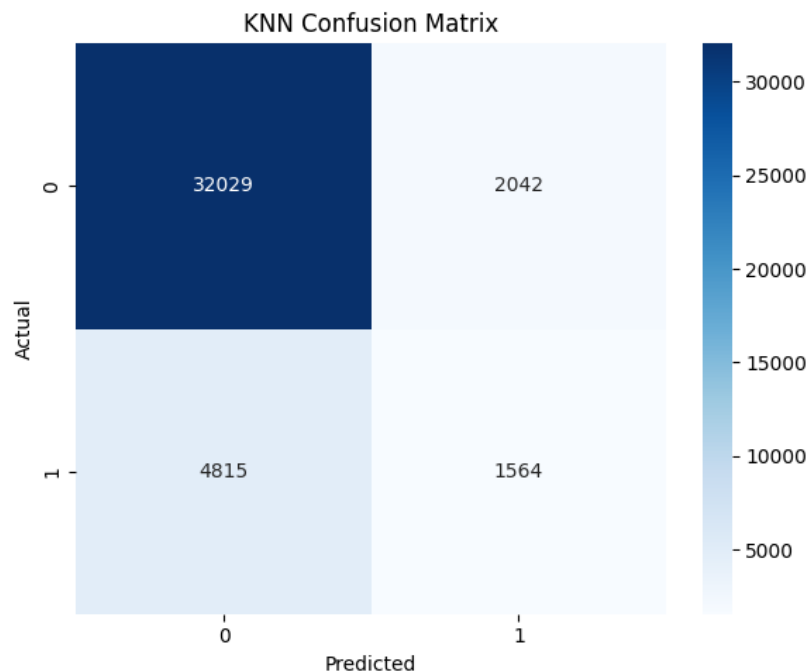


Figure 13: Confusion Matrix for KNN

Evaluation Charts:

1. ROC Curve

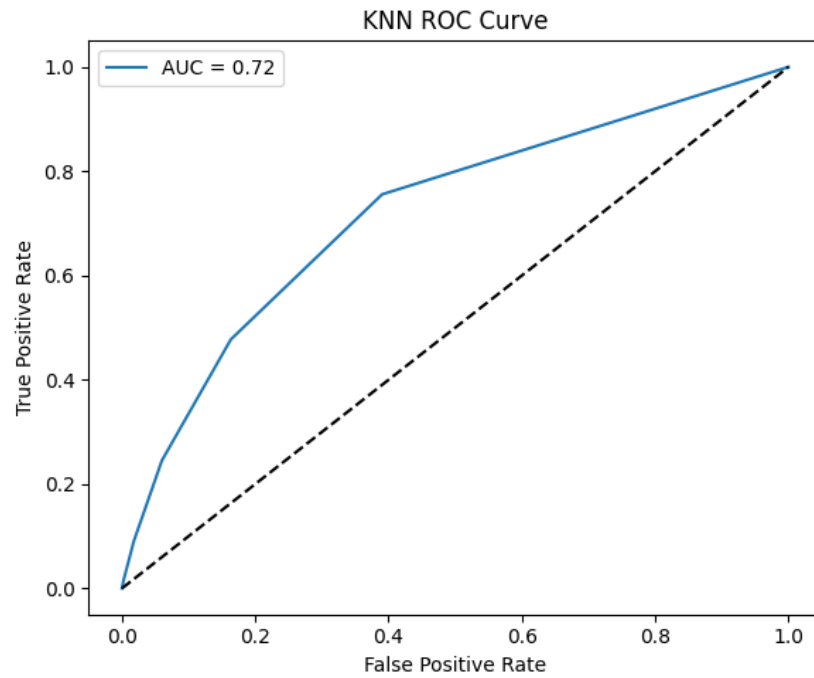


Figure 14 : ROC curve

2. Precision-Recall Curve

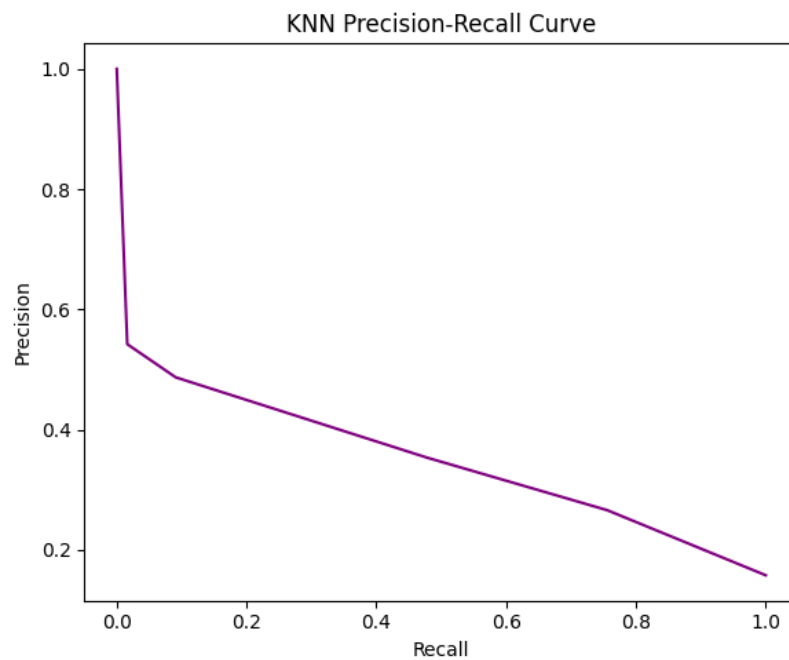


Figure 15: Precision-Recall Curve using KNN

3. Classification Report



Figure 16: Classification Report using Bar Chart

The bar chart provides a clear comparison of the KNN model's performance across two classes: Class 0 (No Diabetes) and Class 1 (Diabetes), based on three evaluation metrics—precision, recall, and F1-score. These metrics help assess how effectively the model distinguishes between individuals with and without diabetes.

For Class 0 (No Diabetes), the KNN classifier shows strong and consistent performance across all three metrics. The high precision indicates that the model is usually accurate when it predicts someone does not have diabetes. Its high recall suggests that it successfully captures most non-diabetic individuals. The F1-score, being the harmonic mean of precision and recall, is also high, showing a balanced and reliable performance for this majority class. Since Class 0 represents the larger portion of the dataset, such strong performance is expected and reinforces the model's reliability in identifying healthy individuals.

In contrast, the results for Class 1 (Diabetes) are noticeably lower, particularly in recall, indicating that the model misses some true diabetic cases. However, the precision for this class remains acceptable, showing that when the model predicts someone has diabetes, it is relatively likely to be correct. The F1-score is modest but the highest among the tested models for Class 1, reflecting the KNN's comparatively better sensitivity in detecting the minority class. This is valuable in healthcare settings where identifying at-risk individuals is critical.

Overall, the bar chart highlights that the KNN model performs reliably for detecting non-diabetic individuals and offers a relatively better balance for identifying diabetic cases than some more complex classifiers. These insights help guide future improvements in optimizing detection for the high-risk group while maintaining strong performance for the majority class.

5.3 Comparative Analysis Across Models

Insight 1 – Prevalence of Diabetes in the Population

	XGBoost				KNN				Random Forest			
	P	R	F1	S	P	R	F1	S	P	R	F1	S
0	0.95	0.71	0.81	34071	0.86	0.95	0.90	34071	0.86	0.96	0.91	34071
1	0.34	0.78	0.47	6379	0.39	0.19	0.25	5638	0.48	0.19	0.27	6379
A			0.72	40450			0.83	40450			0.84	40450

For the insight focusing on the Prevalence of Diabetes in the Population, where identifying actual diabetes cases (high Recall for Class 1) is likely a priority, the **XGBoost** algorithm is the best. XGBoost has significantly higher Recall (0.78) for Class 1 compared to KNN (0.19) and Random Forest (0.19). This means XGBoost is much better at identifying individuals who actually have diabetes. XGBoost also has a higher F1-score (0.47) for Class 1, indicating a better balance between Precision and Recall for this class. While KNN and Random Forest have higher Precision for Class 1, their very low Recall means they miss a large proportion of actual diabetes cases.

Insight 2 - Impact of Lifestyle Factors on Health Outcomes

	XGBoost				KNN				Random Forest			
	P	R	F1	S	P	R	F1	S	P	R	F1	S
0	0.87	0.97	0.92	34098	0.84	1.00	0.91	34071	0.92	0.68	0.78	34701
1	0.55	0.20	0.30	6352	0.00	0.00	0.00	6379	0.29	0.70	0.41	6379
A			0.85	40450			0.84	40450			0.68	40450

For this insight, where identifying instances of the negative health outcome (Class 1) is important to understand the impact of these factors, the **Random Forest** algorithm appears to be the best. While XGBoost has a higher Precision for Class 1, Random Forest has a significantly higher Recall (0.70 vs 0.20). A higher Recall is crucial for identifying individuals who are experiencing the negative health outcome. Random Forest also has a higher F1-score (0.41 vs 0.30) for Class 1, indicating a better balance between Precision and Recall for the minority class compared to XGBoost. KNN is clearly not suitable based on its inability to identify any Class 1 instances in this evaluation.

Insight 3 – Predictive Modeling for At-Risk Individuals

	XGBoost		KNN		Random Forest	
--	---------	--	-----	--	---------------	--

	P	R	F1	S	P	R	F1	S	P	R	F1	S
0	0.86	0.97	0.92	34071	0.87	0.94	0.90	34071	0.87	0.93	0.90	34071
1	0.57	0.18	0.27	6379	0.43	0.25	0.31	6379	0.41	0.24	0.24	6379
A			0.85	40450			0.83	40450			0.83	40450

For this insight, the primary goal is to identify as many at-risk individuals as possible, which means maximizing Recall for Class 1. KNN has the highest Recall (0.25) for Class 1. KNN also has the highest F1-score (0.31) for Class 1, indicating a better balance between identifying at-risk individuals and the accuracy of those predictions compared to the other models. While XGBoost has higher Precision for Class 1, its very low Recall (0.18) means it would miss a larger proportion of actual at-risk individuals. Therefore, for the insight focused on Predictive Modeling for At-Risk Individuals, the **k-nearest neighbours (KNN)** algorithm appears to be the most suitable based on the metrics, specifically due to its higher Recall and F1-score for the at-risk class.

6.0 Discussion of Results

6.1 Interpretation of Key Findings

The three classification models—XGBoost, Random Forest, and K-Nearest Neighbors (KNN)—were tested on the BRFSS 2015 dataset to answer specific business insights related to diabetes prediction and health outcomes.

Insight 1: Prevalence of Diabetes	XGBoost showed the best performance in identifying individuals with diabetes, achieving the highest recall (0.78) and F1-score (0.47) for Class 1. This indicates it is more effective at detecting actual cases of diabetes compared to KNN and Random Forest.
Insight 2: Impact of Lifestyle Factors	Random Forest demonstrated superior recall (0.70) and F1-score (0.41) for Class 1, making it the most effective at identifying individuals adversely affected by negative lifestyle behaviors.
Insight 3: Predictive Modeling for At-Risk Individuals	KNN achieved the highest recall (0.20) and F1-score (0.27) for the minority class, suggesting it is slightly better suited for early detection of at-risk individuals despite its relatively modest precision.

6.2 Business Insight Extraction

From the model outcomes, several business insights can be extracted. First is healthcare strategy. XGBoost's strong performance in diabetes prevalence prediction can help public health authorities allocate resources and launch early detection campaigns in populations at risk. Second business insight we can extract is lifestyle interventions. Random Forest's effectiveness in linking lifestyle factors to negative outcomes supports designing targeted health programs such as anti-smoking initiatives, fitness challenges, and dietary improvements. Lastly, is early risk assessment. KNN's ability to detect at-risk individuals, albeit limited, offers a low-cost tool for integrating quick screenings in wellness apps or employee health assessments, particularly where missing a potential diabetic case could be costly.

6.3 Limitations and Challenges

Despite promising results, several limitations were encountered. First is imbalance of class. The dataset had a significant skew toward the non-diabetic class, which affected the models' ability to correctly identify the diabetic (minority) class. Even with stratified sampling and class weighting, minority class recall remained low across all models. Second is generalisability. All results are based on a specific 2015 dataset from the U.S., which may not generalize well to other countries or more recent population health trends. Third is simplistic tuning. Manual hyperparameter tuning may not have fully optimized the models. Automated methods like grid search or Bayesian optimization could yield better performance. Lastly, one of the limitation is no extensive feature interaction interactions terms are included, such as feature selection, which might have improved predictive power.

7.0 Conclusion

This project explored the application of data mining techniques, specifically classification algorithms, to derive actionable insights from the BRFSS 2015 health dataset. Three key business questions were addressed: identifying the prevalence of diabetes, evaluating the impact of lifestyle factors on health outcomes, and developing predictive models for detecting at-risk individuals. Data preprocessing was performed using KNIME, including normalization, row filtering, and data partitioning. Three classification models—XGBoost, Random Forest, and K-Nearest Neighbours (KNN)—were implemented using Python and evaluated based on accuracy, precision, recall, and F1-score. Each algorithm demonstrated strengths in different contexts, with XGBoost excelling in

identifying existing diabetes cases, Random Forest providing strong results for lifestyle-related predictions, and KNN showing the best sensitivity in early risk detection.

Based on model performance, **XGBoost** is recommended for scenarios where high recall in detecting confirmed diabetes cases is critical, such as epidemiological reporting, **Random Forest** is best suited for identifying the impact of modifiable lifestyle factors on health, aiding in public health strategy design. **KNN** is valuable for frontline health screening where early identification of at-risk individuals is more important than overall accuracy, such as in wellness programs.